

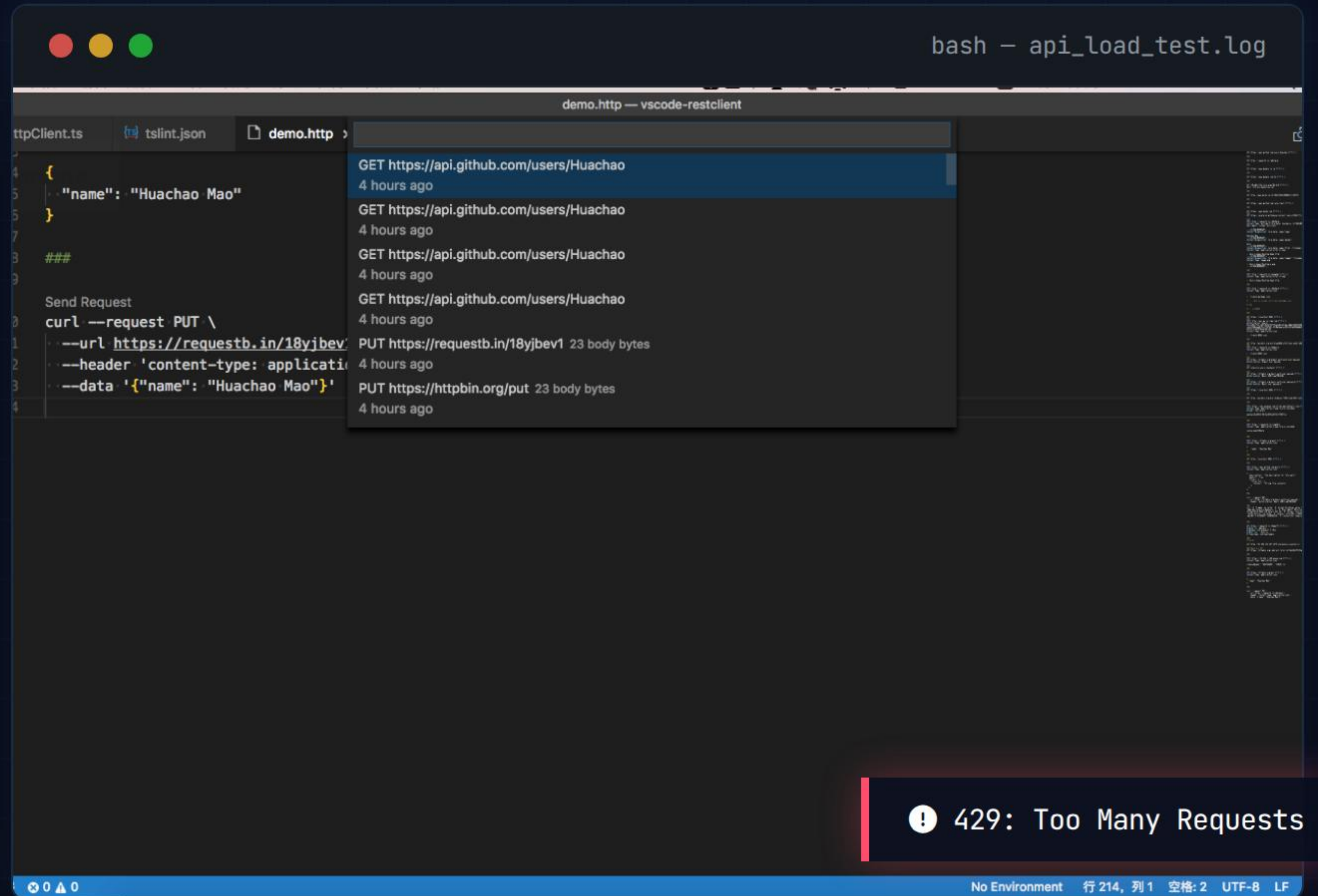
# Distributed API Rate Limiter

THRESHOLD ENFORCEMENT: 10 ALLOWED / 5 THROTTLED

Ensuring system stability by enforcing request quotas across shared state.

 **Shared Persistence**  
Leveraging Redis for global request tracking across nodes.

 **HTTP 429 Protection**  
Automated back-off signals sent to clients on quota breach.



```
bash - api_load_test.log

demo.http - vscode-restclient

4 {
5   "name": "Huachao Mao"
6 }
7
8 ###
9
10 Send Request
11 curl --request PUT \
12   --url https://requestb.in/18yjbev1 \
13   --header 'content-type: application/json' \
14   --data '{"name": "Huachao Mao"}'

GET https://api.github.com/users/Huachao
4 hours ago
GET https://api.github.com/users/Huachao
4 hours ago
GET https://api.github.com/users/Huachao
4 hours ago
GET https://api.github.com/users/Huachao
4 hours ago
GET https://api.github.com/users/Huachao
4 hours ago
PUT https://requestb.in/18yjbev1 23 body bytes
4 hours ago
PUT https://httpbin.org/put 23 body bytes
4 hours ago
```

429: Too Many Requests

Rate limiting successfully enforced using Redis shared state.



# Production Headers

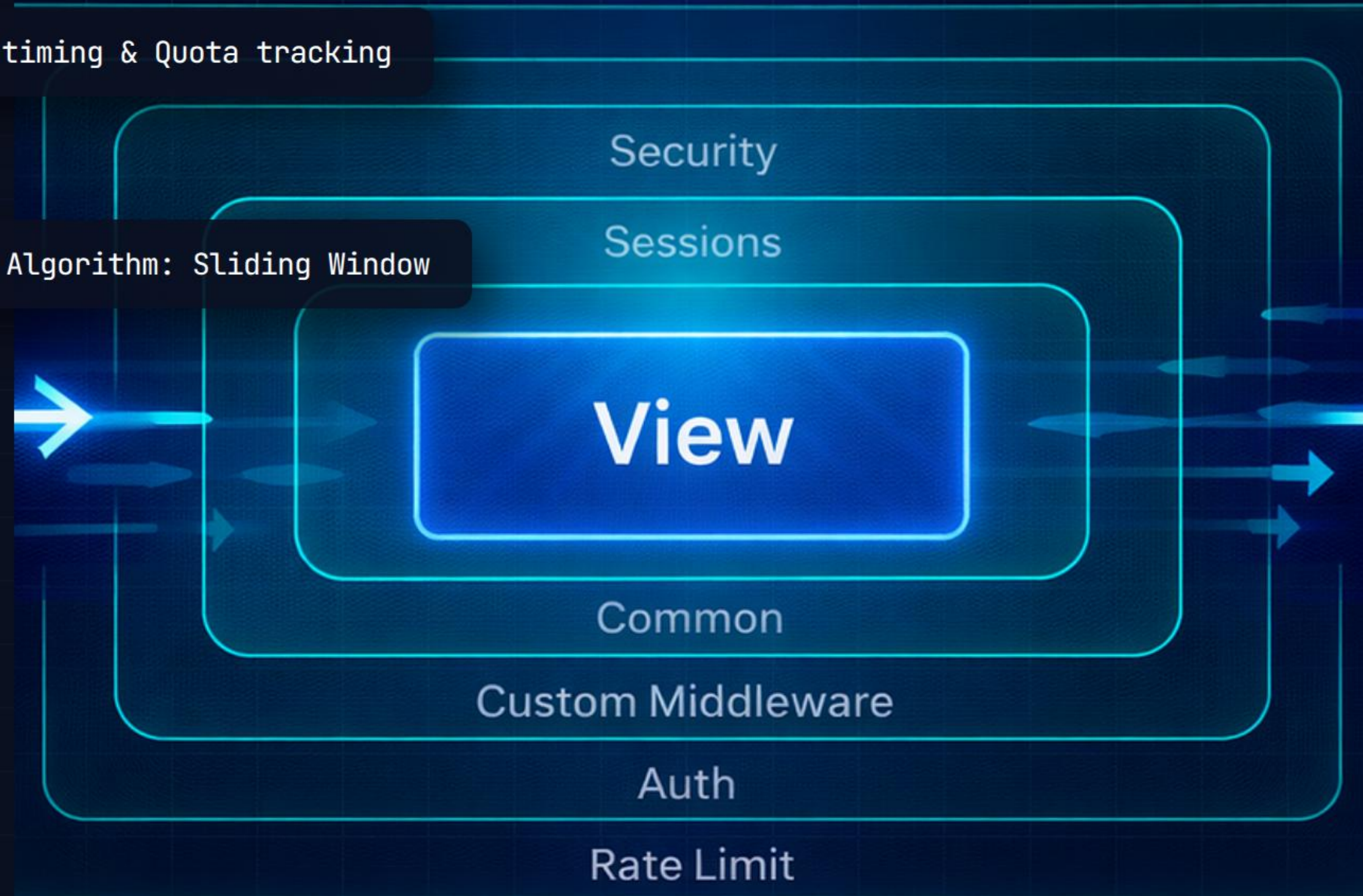
## RETRY TIMING & METADATA

Communicating rate limit status via standardized HTTP headers for efficient client-side orchestration.

- ▶ X-RateLimit-Limit
- ▶ X-RateLimit-Remaining
- ▶ Retry-After: 45s

🕒 Retry timing & Quota tracking

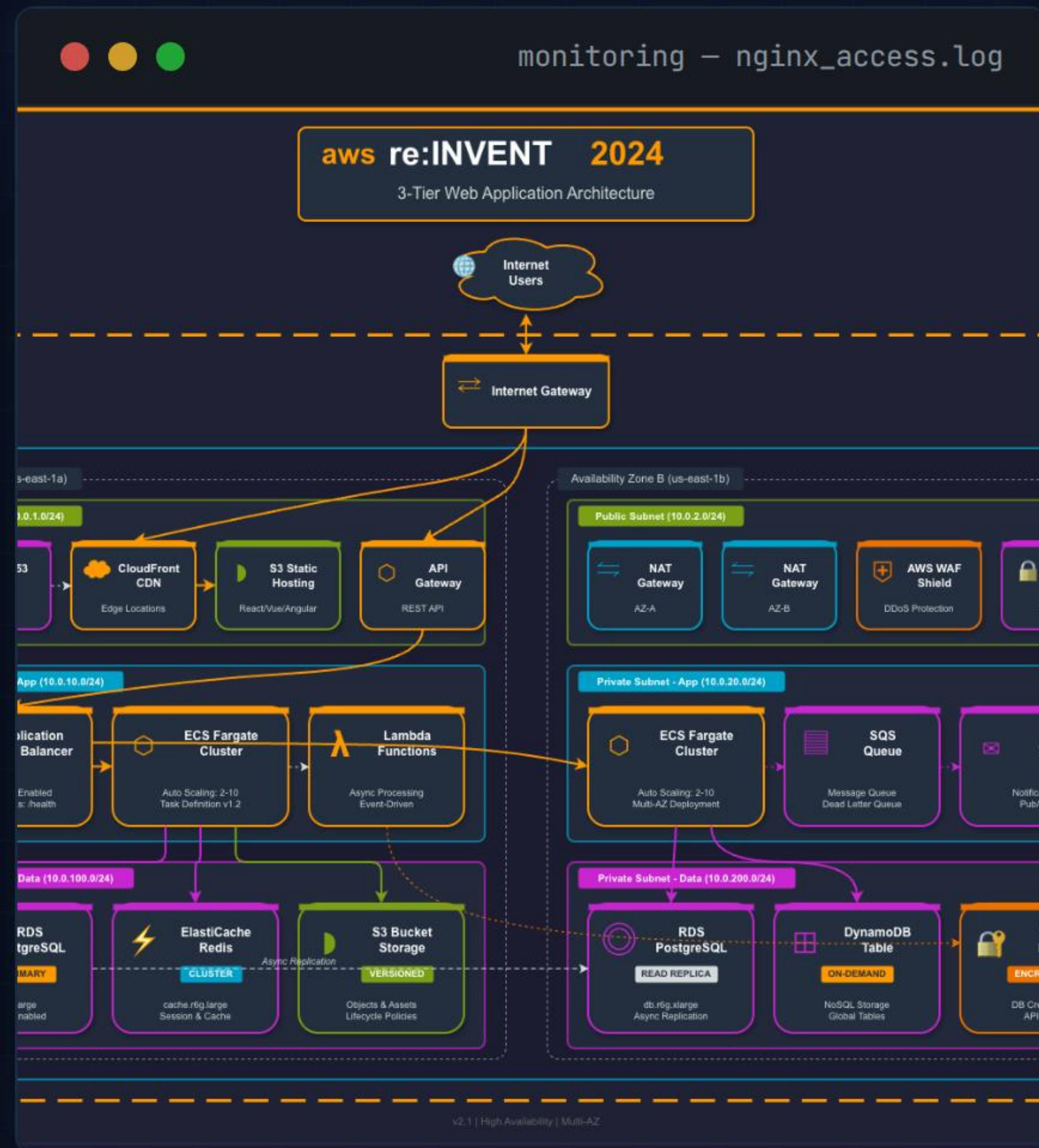
📖 Algorithm: Sliding Window





# Distributed & Load Balanced

## NGINX TRAFFIC DISTRIBUTION



```
[proxy_pass] → api_upstream_cluster
```

Node A: e99b4c7bf9c8

Node B: 7b3e2424f9cb

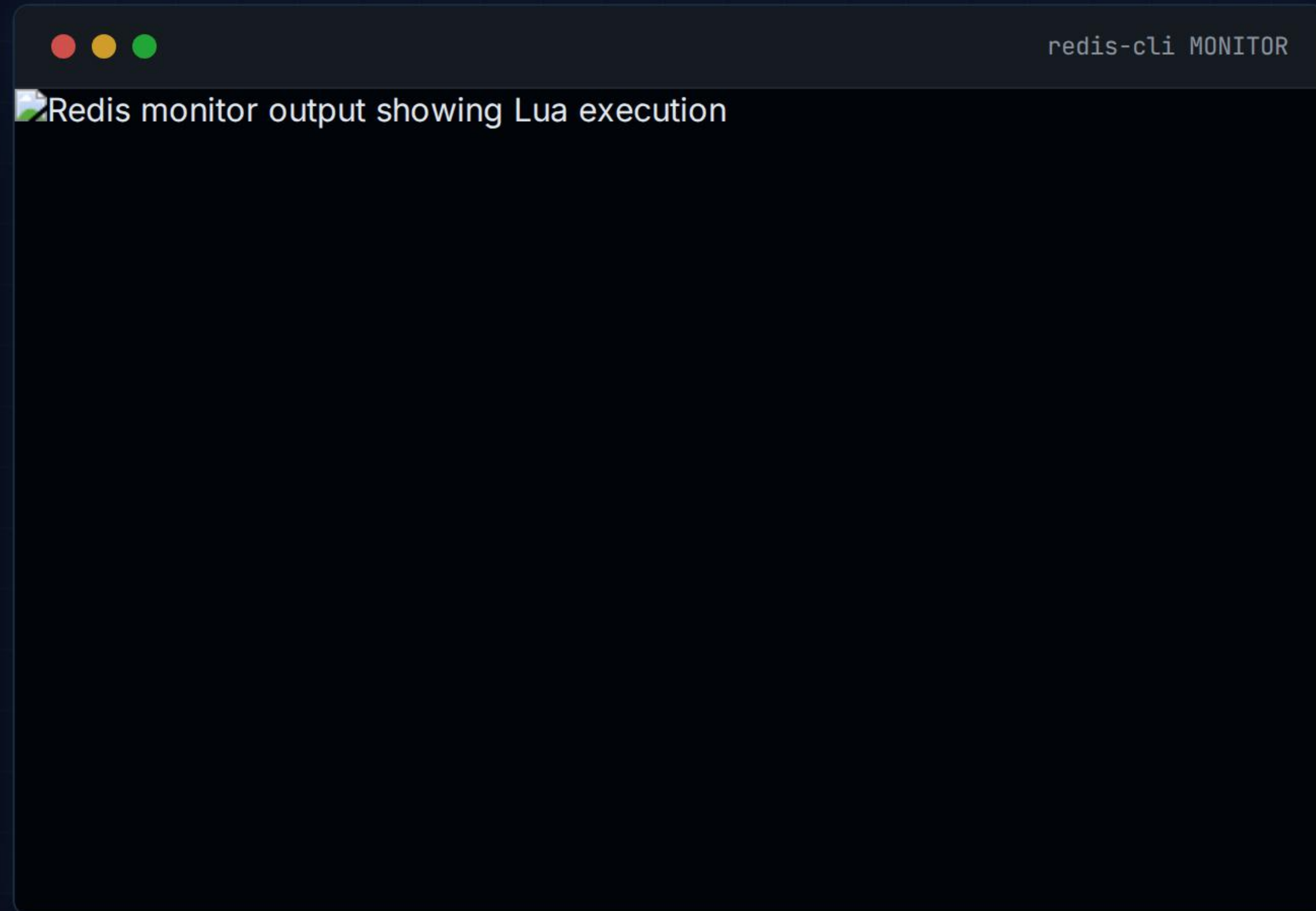
```
# Round Robin balancing enabled
```

Requests are distributed across multiple API instances to ensure high availability and horizontal scaling.



# Redis Lua Scripts (Atomic)

PREVENTING RACE CONDITIONS ACROSS INSTANCES



```
Infra logic: EVAL "local c = redis.call('incr', KEYS[1]) ..."
```



## Atomic EVAL

Scripts are executed as a single unit, preventing interleaved commands from concurrent requests.



## Logic offloading

Check quota, increment, and set TTL in a single network round-trip.



Container memory usage ⓘ  
83.95MB / 7.48GB

Only show running containers

| Ima...                | ↑ | Status  | CPU... | Port(s)                 | Last started   |
|-----------------------|---|---------|--------|-------------------------|----------------|
| <a href="#">nginx</a> |   | Running | 0%     | <a href="#">80:80</a> ↗ | 44 seconds ago |
| <a href="#">redis</a> |   | Running | 0.3%   |                         | 1 minute ago   |
| <a href="#">web</a>   |   | Running | 0.29%  |                         | 1 minute ago   |
| <a href="#">web</a>   |   | Running | 0%     |                         | 57 seconds ago |

System Orchestrated

# Containerized Infra

DOCKER COMPOSE ORCHESTRATION

Unified environment for local development and staging replication.

- /nginx-lb — External Gateway
- /api-nodes — Horizontal Cluster
- /redis-shared — State Store



# Image Sources



<https://raw.githubusercontent.com/Huachao/vscode-restclient/master/images/request-history.png>

Source: [marketplace.visualstudio.com](https://marketplace.visualstudio.com)

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[https://miro.medium.com/v2/resize:fit:1400/1\\*RNio4hrRgaNciWBaiEJ8dw.png](https://miro.medium.com/v2/resize:fit:1400/1*RNio4hrRgaNciWBaiEJ8dw.png)

Source: [levelup.gitconnected.com](https://levelup.gitconnected.com)

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<https://robotpaper.ai/content/images/2026/02/image-7.png>

Source: [robotpaper.ai](https://robotpaper.ai)

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<https://i.ytimg.com/vi/IMvRfStaoyM/maxresdefault.jpg>

Source: [www.youtube.com](https://www.youtube.com)

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A screenshot of the Docker Desktop interface showing a list of running containers. The table has columns for Name, Status, CPU, Memory, and Last State. The containers are listed as 'nginx', 'nginx', 'nginx', and 'nginx'.

<https://docs.docker.com/get-started/docker-concepts/running-containers/images/multi-container-apps.webp>

Source: [docs.docker.com](https://docs.docker.com)